

INVERTEBRATES

Invertebrates are animals without backbones. They account for 97 percent of all animal species, almost covering the full range of animal diversity—from brainless jellyfish to flying insects. Beetle species alone may outnumber vertebrates ten to one. Invertebrates dominate by individuals too: at any one time, the

ocean's shrimplike krill outnumber all the humans who have ever lived; and a single super-colony of ants can contain billions. Some invertebrates even occupy habitats too hostile for other life forms—including around boiling hot vents on the ocean floor or in lakes of hypersaline brine.

Anatomy

Invertebrates are defined by exclusion. They include all the species that are not classified as vertebrate (back-boned) animals, thus covering more than 30 phyla.

The lowliest of invertebrates—shapeless sponges—lack all bodily organs. The rest are muscular, animate creatures that can swim, burrow, walk, or fly. Jellyfish and anemones have a radial anatomy. Their tentacles encircle a single gut opening, so—anatomically—they have “up” and “down” but no “front” and “back.” Most other animals have a “head end” that leads forward—with a brain that can coordinate the sensory information received from the surroundings.

The biggest range of anatomical types among invertebrates occurs in the world's oceans, including animals that burrow in sediments or swim in open water, but groups from different phyla have invaded land too. On land, invertebrates need an especially effective skeleton for support—as well as protection from dehydration. Arthropods—jointed-legged invertebrates, such as insects and spiders—have their cuticle reinforced into a rigid exoskeleton. It makes them true conquerors of the planet—they now make up more than 80 percent of the world's animal life, the articulated parts of their jointed legs giving them superior mobility in running, digging, and swimming.

Reproduction

Some invertebrates can reproduce without sex: sponges and some flatworms can fragment, while aphids give birth to young without being fertilized. However, sexual reproduction, where sperm cells fuse with eggs, is more widespread because it has the benefit of mixing up genetic variety. It can be astonishingly productive.

Minute, anemonelike polyps of a coral reef eject billions of eggs and sperm at once, with external fertilization happening in open water. Other invertebrates transfer sperm directly into the female's body. This kind of internal fertilization is crucial on land but demands complex sexual parts. Slugs, for instance, jab their partners with “love darts” to facilitate sperm transfer.

Some invertebrates may live only for a few hours, but some ocean clams survive for centuries. The life of many invertebrates is punctuated by episodes of metamorphosis too, whereby larval forms undergo considerable reshaping as they develop into adults. For example, many kinds of marine worms, snails, and crustaceans start their lives as minute planktonic larvae.

Behavior

Invertebrates have varied and impressive sensory adaptations. Male moths can sense just a few molecules of sexual pheromone wafting from a



◁ GROWING UP

Larval invertebrate stages may be different from their adult forms. The mullein moth caterpillar is a colorful leaf-eater before metamorphosing into a drab, nectar-drinking adult.

female miles away, while predatory mantis shrimps see sun-dappled coral reefs using highly sophisticated color vision.

However, invertebrates generally have simple brains for habitual tasks—for example, a fairy wasp, which is the size of a period, has a brain with 10 billion times fewer neurons than that of a human. The wasp's behavior is genetically “hard-wired” and inflexible although still complex enough to parasitize the eggs of other insects. Many invertebrates—such as ants, termites, and bees and wasps—are highly social and exhibit a bigger behavioral repertoire by living in complex societies with division of labor. These colonies are made up of different “caste” members devoted to foraging, defense, and breeding—effectively meaning that the entire colony works more like a super-organism with a “super-brain.”

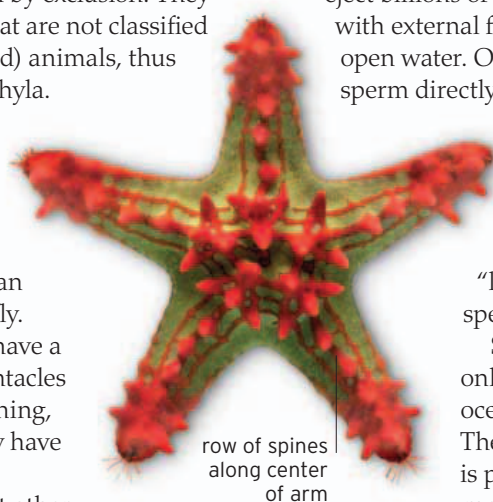


◁ PREDATORY ACT

Although hard-wired, invertebrate behavior is complex. A spider-hunting hawk wasp drags its victim—paralyzed by stinging—to a nest to provide food for its larva.

▷ SUPERSIZED

The widest variety of invertebrate phyla is oceanic. Many are tiny, wormlike animals, but others—such as this octopus—grow large and exhibit complex behavior.



△ RADIAL SYMMETRY

Most invertebrates have an elongated body, running from head to “tail.” However, some—such as starfish—have a radial body encircling a central gut cavity.